

**LINUX
SCENARIO
BASED Q&A
DAY 6**

Q1. Boot Fails: "No init Found" or Systemd Missing

🌿 Scenario: System fails to boot with an error like "No init found", or stalls during early boot stages.

🧠 Possible Root Cause:

- The init process (/sbin/init or /lib/systemd/systemd) is missing, broken, or was accidentally modified.

🔧 Solution:

1. Boot from a Live ISO or rescue environment.
2. Mount the root filesystem and chroot into it:
`sudo chroot /mnt`
`sudo apt install --reinstall systemd`

✅ Tip:

Never delete or tamper with /sbin/init or systemd binaries manually, they are essential for the Linux boot process.

Q2. New Disk Not Mounted on Boot

🌿 Scenario: You've attached a new disk, but it isn't mounted automatically after reboot.

🧠 Possible Root Cause:

- The disk is missing from /etc/fstab, or the system skips it during boot due to delays or initialization timing.

🔧 Solution:

1. Add the disk to /etc/fstab using the nofail option to prevent boot failure:
`UUID=xxxx-xxxx /mnt/data ext4 defaults,nofail 0 2`
2. Test the entry:
`sudo mount -a`

✅ Tip:

Use the nofail option to ensure the system still boots even if the disk is temporarily unavailable.

Q3. SELinux Prevents Services from Starting After Reboot

🌿 Scenario: After a reboot, services like web or database servers fail to start. Logs show "permission denied" errors.

🧠 Possible Root Cause:

- SELinux is blocking access due to incorrect security contexts on service files or directories.
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🔧 Solution:

1. Temporarily set SELinux to permissive mode for troubleshooting:
`sudo setenforce 0`
2. Restore correct file contexts (e.g., for web root):
`sudo restorecon -Rv /var/www`

✅ Tip:

Use audit2allow to analyze denial logs and generate custom SELinux policy modules if persistent access is needed.

Q4. YUM Fails: "Cannot find a valid baseurl"

✿ Scenario: When running yum install, it fails with:
Cannot find a valid baseurl for repo

🧠 Possible Root Cause:

- DNS resolution failure
- Incorrect or outdated repo URLs
- No internet connectivity

🔧 Solution:

1. Verify internet access:
ping google.com
2. Inspect repo configuration:
cat /etc/yum.repos.d/*.repo
Update or correct the baseurl as needed.

✅ Tip:

For legacy CentOS or RHEL systems, consider using CentOS Vault or EPEL repositories to restore functionality.

Q5. Install Fails: "Package is already installed"

✿ Scenario: Attempting to install a package results in:
dpkg: error processing package -- already installed

🧠 Possible Root Cause:

- A previous install was interrupted or left the package in a half-configured state.

🔧 Solution:

1. Reconfigure and repair broken installs:
sudo dpkg --configure -a
sudo apt install -f

✅ Tip:

This often happens if you cancel an install (e.g., with CTRL+C). Always let apt or dpkg complete to avoid inconsistent states.

Q6. RPM Install Fails: "Failed Dependencies"

✿ Scenario: Installing an .rpm package throws errors about missing libraries or dependencies.

🧠 Possible Root Cause:

- Required dependencies are not installed or not found in enabled repositories.
- Using rpm -i instead of a dependency-resolving tool like yum or dnf.

🔧 Solution:

1. Use YUM to install with dependencies: sudo yum localinstall package.rpm
2. Or use DNF for modern systems: sudo dnf install ./package.rpm

✅ Tip:

Use yum provides <missing-file> to identify which package contains a specific missing dependency.

Q7. apt install Installs an Older Package Version

✚ Scenario: When installing a package, APT installs an outdated version instead of the latest.

🧠 Possible Root Cause:

- Package lists are outdated (apt update not run)
- Latest version isn't available in the current repositories

🔧 Solution:

1. Refresh APT cache:

```
sudo apt update
```

2. Install a specific version manually:

```
sudo apt install pkgname=1.2.3-1
```

✅ Tip:

Use this to check available versions:

```
apt policy pkgname
```

Q8. Installing One Package Breaks Another

✚ Scenario: Installing a new package causes an existing application to downgrade or malfunction.

🧠 Possible Root Cause:

- Conflicting versions across repositories
- Manual .deb or .rpm install overrides system-managed versions

🔧 Solution:

1. For Debian/Ubuntu – Use APT Pinning:

Lock a specific version to prevent unwanted changes:

```
echo -e "Package: pkgname\nPin: version 1.2.3\nPin-Priority: 1001" | sudo tee /etc/apt/preferences.d/pkgname
```

2. For RHEL/CentOS – Use Version Lock:

```
yum versionlock add pkgname
```

✅ Tip:

Pin or lock critical packages before introducing external or manually downloaded packages.

Q9. How to Downgrade a Package Cleanly

✚ Scenario:

A newly installed version causes issues, and you need to revert to a previous stable release.

🧠 Possible Root Cause:

Package managers like APT or YUM/DNF don't automatically support downgrades unless explicitly instructed.

 Solution:

1. For APT-based systems:
`sudo apt install package=old_version`
2. For DNF-based systems (RHEL/Fedora):
`sudo dnf downgrade package`

 Tip:

To prevent the package from upgrading again, pin or lock the version:

- APT: Use `/etc/apt/preferences.d/`
- YUM/DNF: Use `yum versionlock add package`

Q10. Package Installed, but “Command Not Found” Error

 Scenario: apt confirms the package is installed, but running the command returns: command not found.

 Possible Root Cause:

- The binary isn't located in a standard `$PATH` directory
- Some tools require a login shell to expose their path
- Installed package might only provide libraries or utilities not meant to be run directly

 Solution:

1. Locate the binary manually:
`dpkg -L pkgname | grep /bin`
2. If needed, create a symlink to expose it globally:
`sudo ln -s /opt/tool/bin/exe /usr/local/bin/exe`

 Tip:

Use `which`, `type -a`, or `echo $PATH` to check command visibility and ensure your shell can see the executable.